

KATHIRVEL A

Firmware Developer

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kathirvel-25.github.io/Portfolio/

PROFESSIONAL SUMMARY

Aspiring Firmware Engineer with hands-on experience in STM32, ESP32, and bare-metal Embedded C. Skilled in low-level driver development, peripheral interfacing, and embedded system design, seeking opportunities in firmware development and embedded systems engineering.

SKILLS

Programming Languages - C, C++, Embedded C

Embedded Concepts - Bare-metal Programming, Interrupt Handling

Communication Protocols - I2C, UART, SPI

MCU Peripherals – ADC, Timer, PWM

Real-Time System – FreeRTOS

Microcontrollers – STM32(ARM Cortex-M4), ESP32, Arduino

Tools – STM32CubeIDE, Arduino IDE, Fusion 360, STM32CubeMX

Robotics and Middleware - Robot Operating System (Learning)

Operating Systems – Linux (Ubuntu)

Debugging Tool – ST-link, Multimeter, RPS

INTERNSHIP EXPERIENCE

Elamani Tech Pvt. Ltd

Mar 2026 – June 2026

Firmware Developer Intern | R&D Department

- Reverse-engineered a commercial delivery robot's embedded hardware architecture, identifying MCU peripherals, sensor interfaces, communication buses, and control mechanisms used in the system.
- Project – Car Lupa**, Developed a bare-metal STM32-based robotic platform using Embedded C, focusing on low-level driver development, peripheral interfacing, and system integration.

PROJECTS

DataLogger using STM32

- Tech Stack:** STM32, Embedded C, UART, SPI Flash, GPS
- Description:** Developed a bare-metal STM32 GPS data logger with interrupt-driven UART communication for real-time GPS data acquisition and serial monitoring. Implemented NMEA data parsing and persistent storage in external SPI Flash memory using custom low-level drivers for reliable embedded data logging.

AI Vision-Based Wireless Robot

- Tech Stack:** Python, MediaPipe, OpenCV, Arduino, HC-05
- Description:** Developed an AI-powered hand-tracking robotic wheelchair that uses MediaPipe and computer vision to recognize real-time hand gestures and wirelessly control wheelchair movement via Bluetooth communication, enabling intuitive touch-free navigation.

STM32 FreeRTOS-Based Elevator Controller

- Tech Stack:** STM32F411RE, Embedded C, FreeRTOS, CMSIS-RTOS2, I2C LCD, PWM, EXTI Interrupts
- Description:** Developed a real-time elevator control system using STM32 and FreeRTOS, implementing multi-tasking architecture for floor request handling, motor control, door operation, display management, and safety monitoring. Utilized message queues, event flags, interrupt-driven inputs, and a finite state machine to coordinate elevator movement, door control, emergency handling, and real-time status updates on an LCD display

EDUCATION

B.E. Electronics and Communication Engineering

2023 - 2027 (April)

Government College of Engineering, Salem | CGPA: 7.8

Higher Secondary

2022 - 2023

Lakshmi Matric Higher Secondary School, Manaparai | Percentage: 93.4%

LEADERSHIP & ACHIEVEMENTS

Class Representative, Department of ECE (Second Semester - Present)

Unit Head, National Service Scheme (NSS).

Overall Mini Project Winner, Among all departments in a college-level competition.

1st Place in Infinity Quest and 3rd Place in Neuronumerics, symposium conducted by the Department of Mathematics, St. Joseph's College.